INCH-POUND
MIL-P-70873 (AR)
10 March 1992

MILITARY SPECIFICATION

PRINTED WIRING BOARD ASSEMBLY, 20 MM ENCODER
SPECIFICATION FOR

This specification is approved for use by the U.S. Army Armament, Munitions and Chemical Command and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification establishes the requirements for manufacture, inspection, packaging and acceptance of the Printed Wiring Board Assembly (PWBA), 20mm Encoder, part number 9339642.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander U.S. Army ARDEC, ATTN: SMCAR-BAC-S, Picatinny Arsenal, New Jersey 07806-5000 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 1265

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2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards and handbooks. The following specifications, standards and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

MILITARY

MIL-F-13926	-	Fire Control Materiel Manufacture and Inspection of, General Specification for
MIL-I-45607	-	Inspection Equipment, Acquisition, Maintenance and Disposition of
MIL-W-63150	-	Weapons and Support Materiel, Standard Quality Assurance Provisions for

STANDARDS

MILITARY

MIL-STD-129	-	Marking for Shipment and Storage
MIL-STD-454	-	Standard General Requirements for Electronic Equipment
MIL-STD-810	-	Environmental Test Methods
MIL-STD-1686	-	Electrostatic Discharge Control Program for Protection of Electrical and Electronic Parts, Assemblies and Equipment (Excluding electrically initiated explosive devices)
MIL-STD-1695	-	Environments, Working, Minimum Standards for
MIL-STD-2000	-	Standard Requirements for Soldered Electrical and Electronic Assemblies
MIL-STD-45662	-	Calibration System's Requirements

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HANDBOOK

MILITARY

DoD-HBK-263 - Electrostatic Discharge Control Handbook for Protection of Electrical and Electronic Parts, Assemblies and Equipment (Excluding electrically initiated explosive devices) METRIC

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from: Standardization Documents Order Desk, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

U.S. ARMY ARMAMENT RESEARCH, DEVELOPMENT AND ENGINEERING CENTER.

PRODUCT DRAWINGS (CAGE CODE 19200)

9339642	-	Printed Wiring Board Assy, 20mm Encoder
9339644	-	Schematic Diagram, 20mm Encoder

INSPECTION EQUIPMENT (CAGE CODE 19200)

11835285	-	Test Set, Decoder (Ref.) (6.8)
11749102	-	Semiconductor, Diode, Light Emitting, Type MV3

PACKAGING DATA SHEET (CAGE CODE 19200)

SPI-P9339642	-	Printed Wiring Board Assy, 20mm Encoder

(Copies of other Government documents, drawings and publications required by the contractor in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.2 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document shall take precedence. Nothing in this specification, however, shall supersede

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applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Item description. The Encoder Board contains a microprocessor and associated logic circuitry which produce the coding, word count, message count and timing for the 20mm cannon.

The board also contains special large scale integrated (LSI) circuitry which serves to produce the signals necessary to control the energy of the laser and to fire the laser in synchronism with commands from the microprocessor. In addition, certain accessory and housekeeping functions are served by this special LSI circuitry. Among them are voltage regulation for the logic circuitry, generation of Power-On-Reset (POR) and amplification of the microphone signal for the weapon firing blank rounds.

3.2 First article. When specified (see 6.4), a sample shall be subjected to first article inspection in accordance with 4.4.

3.3 Environmental stress screen (ESS). All assemblies shall be subjected to an environmental stress screen prior to conformal coating consisting of temperature cycling and soaking between the temperatures -35°C and +62.8°C. After exposure to ESS in accordance with 4.5, the characteristics of 3.4 shall not be degraded.

3.4 Characteristics.

3.4.1 Operating conditions. The operating conditions shall be in accordance with Table I, measured under quiescent conditions. For terminal and component designations called out in this specification, see drawing 9339644, Schematic Diagram, 20mm Encoder.

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TABLE I. Operating Conditions.

Parameter	Terminal	Minimum	Nominal	Maximum	Unit
Ambient Temperature		-31.7	25	55	°C
Input (+) Power	1, 2	8	12	13	Vdc
Return (-) Power	31				
Input Current	1, 2			8.0	mA

3.4.2 Performance. Unless otherwise specified, the PWBA shall meet the requirements of this specification under the following conditions:

a.	Temperature	25° ± 10°C (77° ± 18°F)
b.	Relative humidity	50 ± 30 %
c.	Barometric pressure	725 + 50, -75mm Hg (28.5 + 2.0, -3.0 inches Hg)
d.	Input power and loads	See Figure 1

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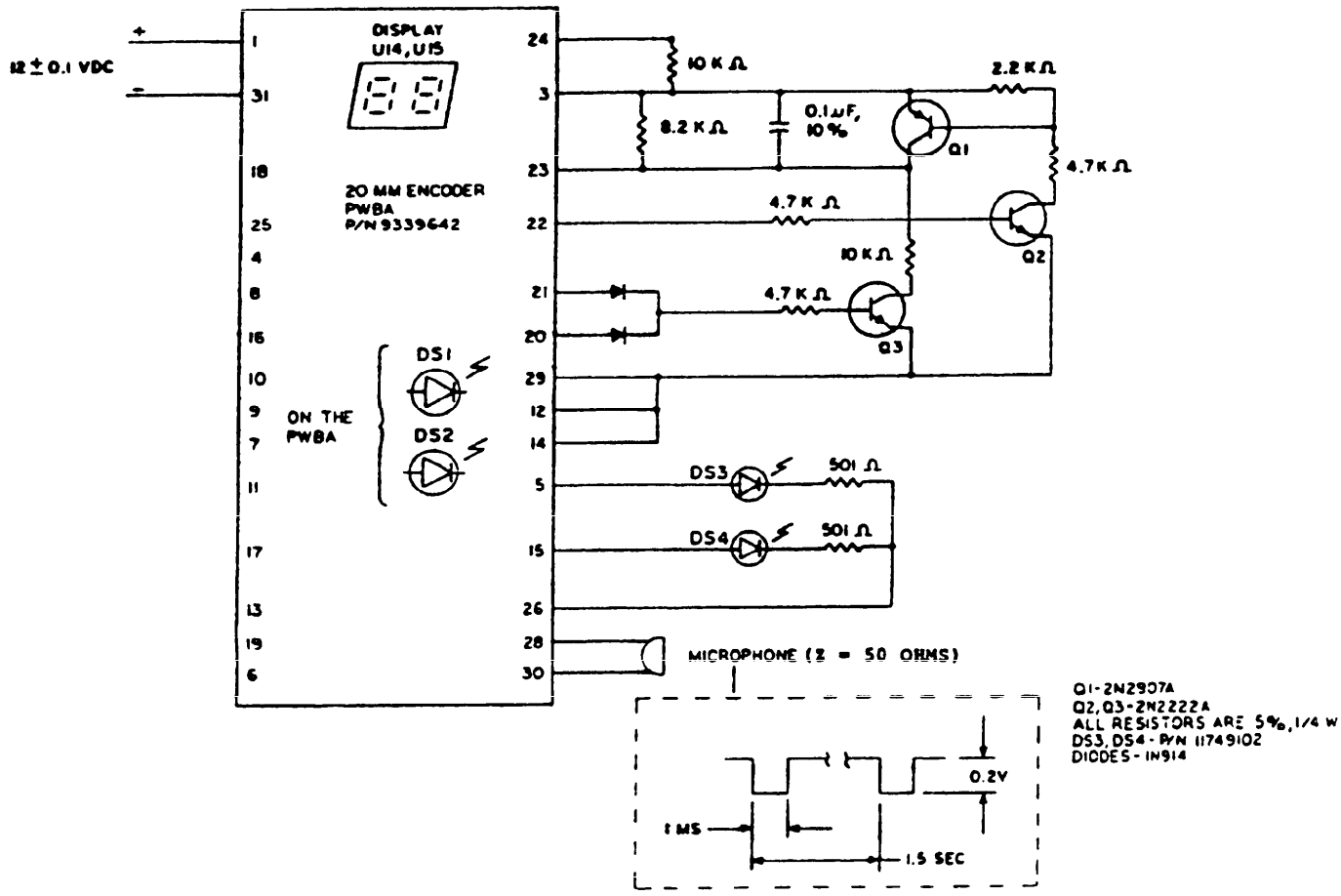


FIGURE 1. 20mm Encoder PWBA Power Source and Loads.

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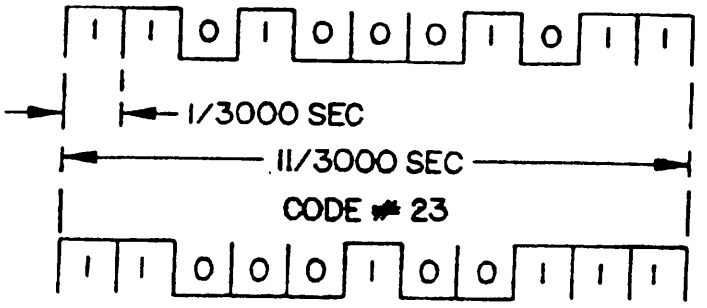
3.4.2.1 Current drain. With power applied to terminals 1(+) and 31 (-), the maximum quiescent current drain shall be 8.0 mA.

3.4.2.2 Rounds count. With power applied, terminal 18 (controller key) grounded momentarily to reset the rounds count, and terminals 17 and 13 disconnected to activate the rounds count display, the display (U14, U15) shall read "75". The controller key signal is sent to a flag of microprocessor U3 which then sends "75" binary coded to the display drivers.

3.4.2.3 Blank fire mode. With power applied, terminals 18 (controller key) and 25 connected momentarily, and the signal shown on Figure 1 applied to terminals 28 (mike in) and 30, apply 28 ± 0.5 V to terminals 7(+) (actual trigger) and 11(-), to trigger the blank fire mode. Then the DS2 LED shall light, and at load terminals 20 and 21, the following codes per round shall appear: 2 copies of Hit code #23, then 19 copies of Miss code #29, then 32 copies of code #23, and then 256 copies of code #29 (see Figure 2). The timing between load terminals 20 and 21 and fire terminals 22 and 24 is shown in Figure 3. LEDs DS1, DS3 and DS4 shall each light every 1.5 sec once the blank fire is triggered. The display (U14, U15) shall initially read "75" and will decrement by 1 for 10 rounds spent every 15 sec once the blank fire is triggered. The microprocessor U3 receives flag signals from the mike in terminal (through the laser driver interface chip U5), the controller key, and the trigger (through relay K1 and a small logic circuit). The microprocessor then sends hit and miss load/fire signals from its I/O lines N0, N1, and N2 to the laser driver chip U4. The DS2 LED lights when the trigger is activated. LEDs DS1, DS3, and DS4 are lighted by microprocessor signals from its I/O lines.

3.4.2.4 Dry fire mode. With power applied, terminals 18 (controller key) and 25 momentarily connected, and terminals 4 (loop) and 8 connected, apply 28 ± 0.5 V to trigger terminals 7(+) and 11(-) for at least 10 milliseconds but less than 60 milliseconds, to trigger the dry fire mode. Then the DS1 and DS3 LEDs shall light during code transmission, the DS4 LED at terminal 15 shall light momentarily, and a single round with the following codes shall appear at load terminals 20 and 21: 2 copies of Hit code #23, then 19 copies of Miss code #29, then 32 copies of code #23, and then 256 copies of code #29 (see Figure 2). The timing between load terminals 20 and 21 and fire terminals 22 and 24 is shown in Figure 3. The microprocessor U3 receives flag signals from the controller key, the loop and the trigger. Hit and miss load/fire signals are then sent from I/O

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CODE #29

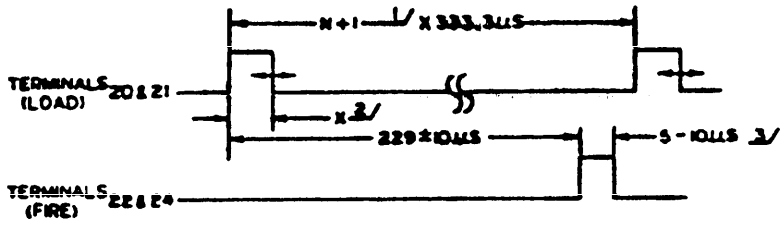
FIGURE 2. Firing Round Codes.

lines N0, N1 and N2 of U3 to the laser driver U4. LEDs DS1, DS3 and DS4 are lighted by microprocessor signals from its I/O lines.

3.4.2.5 Test mode. With power applied, and terminals 18 (controller key) and 25 momentarily connected, initiate the test mode with terminals 16 (test) and 10 momentarily connected. Then LEDs DS1 and DS3 shall each light during code transmission, the resistance between WESS terminals 14 and 15 shall be at least 1 megohm and a single round with the following codes shall appear at load terminals 20 and 21: 2 copies of Hit code #23, then 19 copies of Miss code #29, then 32 copies of code #23 and then 256 copies of code #29 (see Figure 2). The timing between load terminals 20 and 21 and fire terminals 22 and 24 is shown in Figure 3. The test signal deactivates relay K2 causing an open circuit at the WESS terminals, and sends a flag to the microprocessor. Along with the controller key initiated flag, the microprocessor then sends hit and miss load/fire signals with its I/O lines N0, N1 and N2 to the laser driver. LEDs DS1 and DS3 are lighted by microprocessor signals from its I/O lines.

3.4.2.6 Kill inhibit mode. With power applied, terminals 18 (controller key) and 25 momentarily connected, terminals 4 (loop) and 8 connected, 28.0 $\pm$ 0.5V applied to trigger terminals 7(+) and 11(-), lighting LED DS2, and terminals 9 (kill) and 1 connected, to initiate the kill inhibit mode, no codes shall be generated at load terminals 20 and 21, or fire terminals 22 and 24. The microprocessor receives flag signals from the loop, the trigger, the controller key and the kill inputs and will not initiate a response.

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NOTES:
1/ N = INTEGER DEPENDING UPON SUCCESSIVE NUMBER OF ZEROS IN CODE.
2/ X IS VARIABLE TO ADJUST PULSE ENERGY. IN NORMAL OPERATION IT IS A FUNCTION OF SUPPLY VOLTAGE, AMBIENT TEMPERATURE AND COMPONENT TOLERANCES, NORMALLY 6 TO 100 µS.
3/ THERE IS A FIRE PULSE EVERY 333.3µS REGARDLESS OF THE EXISTENCE OF A CORRESPONDING LOAD PULSE.

FIGURE 3. Encoder Board Timing.

3.4.2.7 . Boresight mode. With power applied, terminals 18 (controller key) and 25 momentarily connected, terminals 1 and 9 (kill) connected, terminals 6 and 19 (boresight) connected, and 28.0 ± 0.5 V applied to trigger terminals 7(+) and 11(-), lighting LED DS2, a continuous stream of the Boresight code #32 words shall appear at load terminals 20 and 21 (see Figure 4). The timing between load terminals 20 and 21 and fire terminals 22 and 24 is shown in Figure 3. The microprocessor will receive flag signals from the boresight, the controller key, the kill, and the trigger inputs. The boresight code load/fire signals are then sent by the microprocessor to the laser driver.

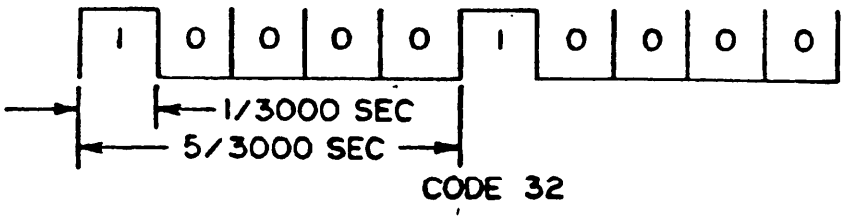


FIGURE 4. Boresight Code.

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3.4.3 Environmental requirements. Unless otherwise specified the assembly shall be subject to the following environmental conditions and meet the stated requirements.

3.4.3.1 High temperature. The assembly shall not be damaged and shall meet the performance requirements of 3.4.2. when exposed to and after the following.

3.4.3.1.1 Operating temperature. The assembly shall operate up to a temperature of +55°C (+130°F).

3.4.3.1.2 Storage temperature. The assembly shall operate after exposure to a temperature of +62.8°C (+145°F) for a period of 48 hours.

3.4.3.2 Low temperature. The assembly shall not be damaged and shall meet the performance requirements of 3.4.2 when exposed to and after the following.

3.4.3.2.1 Operating temperature. The assembly shall operate down to a temperature of -31.7°C (-25°F) after exposure at -31.7°C for 4 hours.

3.4.3.2.2 Storage temperature. The assembly shall operate after exposure to a temperature of -34.4°C (-30°F) for a period of 24 hours.

3.4.3.3 Shock. The assembly shall meet the performance requirements of 3.4.2 after shocks with a nominal level of 40 g's terminal peak and nominal duration of 11 milliseconds as shown in Figure 5. Apply three shocks (18 total) in each direction of the three mutually perpendicular axes.

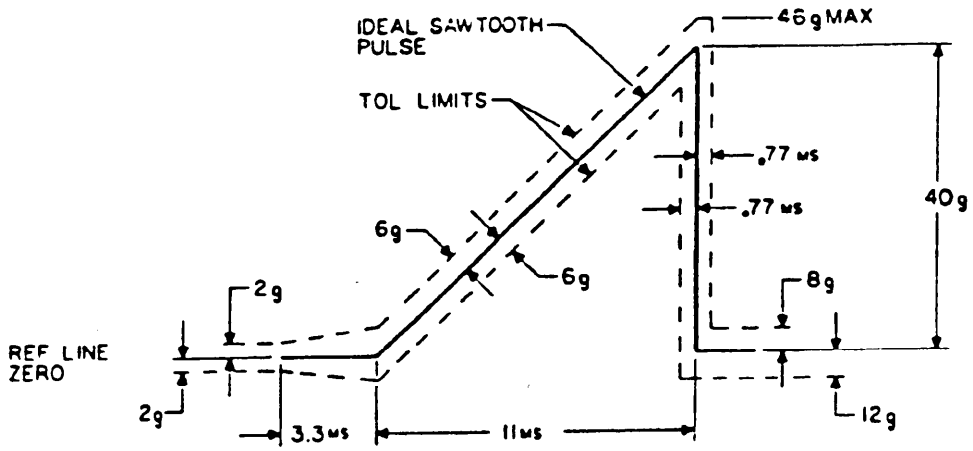


FIGURE 5. Shock Pulse.

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3.4.3.4 Vibration. The assembly shall meet the performance requirements of 3.4.2 after sinusoidal vibrations of frequency and amplitude given below and in Figure 6 on three mutually perpendicular axes. Total duration shall be 60 minutes per axis.

Frequency	Amplitude	Sweep rate
7-30 Hz	±1.5 g	7-500-7 Hz
30-50 Hz	0.033" DA	in 15 minutes
50-500 Hz	±4.2 g	

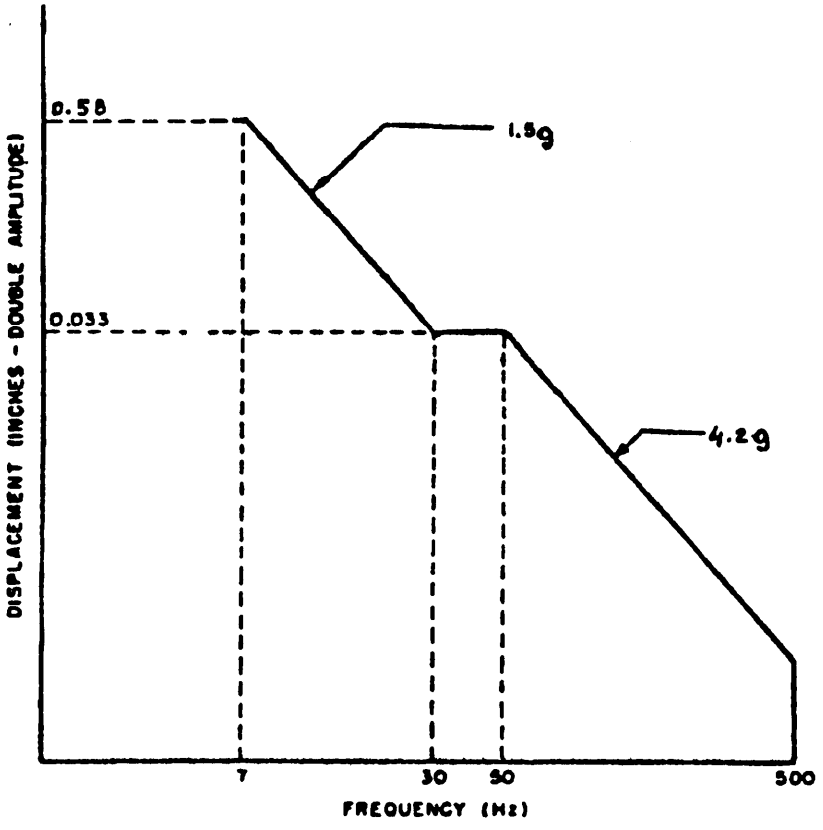


FIGURE 6. Vibration Test.

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3.5 Design and construction. The PWBA shall be fabricated and assembled in accordance with drawings, parts lists, and other documents specified on drawing 9339642. The PWBA shall be manufactured and tested in a working environment in accordance with Table I, work area 5 of MIL-STD-1695.

3.5.1 Standards of manufacture. Handling of Electrostatic Sensitive Parts. This equipment contains semiconductor devices such as CMOS, mosfets, and microelectronic devices which may be degraded or damaged by an inadvertent discharge of static electricity. To prevent damage to these components, the supplier shall provide for adequate handling procedures and equipment in accordance with MIL-STD-1686 and DoD-HBK-263.

3.5.2 Soldering. Soldering shall be in accordance with the general requirements of section 4 of MIL-STD-2000.

3.5.3 Workmanship. General workmanship shall be in accordance with requirement 9 of MIL-STD-454.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

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4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.4).

b. Environmental Stress Screen (see 4.5).

c. Quality conformance inspection (see 4.6).

d. Packaging inspection (see 4.6.4).

4.3 Inspection conditions. Unless otherwise specified, all inspections shall be performed in accordance with the working environment specified in 3.5 and at standard ambient conditions defined in 3.4.2 using the methods of 4.8 and 4.9.

4.4 First article inspection.

4.4.1 Sample. The contractor shall submit a first article sample as designated by the Contracting Officer for evaluation in accordance with provisions of 4.4.2. The first article sample shall consist of three printed wiring board assemblies.

4.4.2 Inspections to be performed. First article inspections are listed in Table II and III. Each first article sample unit shall be inspected in accordance with the Table II before being subjected to the environmental tests in Table III in the sequence listed. As determined by the Government, first article assemblies may be subjected to any or all of the examinations and tests specified in this detail specification and be inspected for compliance with any or all requirements of the applicable drawings.

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TABLE II. Examinations and Performance Tests. 1/ 2/

Class	Characteristics	Requirements Paragraph	Test/ inspection Method	First Article Sample Size	Quality 3/ Conformance Inspection Sample Size/ Accept Criteria
101	Current Drain	3.4.2.1	4.8.1.1	3	32/Acc-0,Rej-1
102	Rounds Count	3.4.2.2	4.8.1.2	3	32/Acc-0,Rej-1
103	Blank Fire Mode	3.4.2.3	4.8.1.3	3	32/Acc-0,Rej-1
104	Dry Fire Mode	3.4.2.4	4.8.1.4	3	32/Acc-0,Rej-1
105	Test Mode	3.4.2.5	4.8.1.5	3	32/Acc-0,Rej-1
106	Kill Inhibit Mode	3.4.2.6	4.8.1.6	3	32/Acc-0,Rej-1
107	Boresight Mode	3.4.2.7	4.8.1.7	3	32/Acc-0,Rej-1
201	Workmanship	3.5.3	Visual	3	125/Acc-0,Rej-1
202	Soldering 4/	3.5.2	Visual	3	125/Acc-0,Rej-1
203	Conformal Coating	3.5	Visual	3	125/Acc-0,Rej-1
204	Dim Characteristics	3.5	SME 5/	3	125/Acc-0,Rej-1
205	Comp Orientation	3.5	Visual	3	125/Acc-0,Rej-1
206	Comp and Board Identification	3.5	Visual	3	125/Acc-0,Rej-1

NOTES:
1/ ESS testing per 4.5 is required for all deliverable units.
2/ 200 class (minor characteristics) examinations required prior to 100 class (major characteristics) performance tests.
3/ For lot sizes less than 32 or 125 as listed, 100% inspection is required (see 4.6.5).
4/ Check before conformal coating is applied.
5/ Standard Measuring Equipment.

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TABLE III. Environmental Tests.

Class	Characteristic	Requirements Paragraph	Test Paragraph	First Article Sample Size	Quality Conformance Inspection Sample Size, Accept Criteria
108	High temp	3.4.3.1	4.9.1	3	1/Acc-0, Rej-1
109	Low temp	3.4.3.2	4.9.2	3	1/Acc-0, Rej-1
110	Shock	3.4.3.3	4.9.3	3	1/
111	Vibration	3.4.3.4	4.9.4	3	1/

NOTE:
1/ Not required for quality conformance inspection.

4.4.3 Rejection. If any assembly fails to comply with any of the applicable requirements, the first article sample shall be rejected. The Government reserves the right to terminate inspection upon any failure of any assembly to comply with any of the requirements.

4.5 Environmental stress screen. An environmental stress screening program shall be established. All units shall be subjected to a thermal stress screen in accordance with Table IV and the procedure below.

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TABLE IV. ESS Temperature Cycling Requirements.

Low Temp	High Temp	Rate Change	Power	Monitor	Cycles/ Group A Test
-34.4°C	+62.8°C	15°C/minute	Off	No	20/Failure Free or Additional Cycle

4.5.1 Conformal coating. Thermal stress screening shall be performed prior to conformal coating unless another method is approved by the Government.

4.5.2 Chamber temperature. Chamber temperature cycling shall be from -34.4°C (-30°F) to +62.8°C (+145°F) and the duration of one cycle to be 8 minutes up and 8 minutes down plus 4 minutes dwell at high temperature and 4 minutes at low temperature for a total cycle duration of 24 minutes. The units shall begin and end the cycles at ambient temperature (see Figure 7).

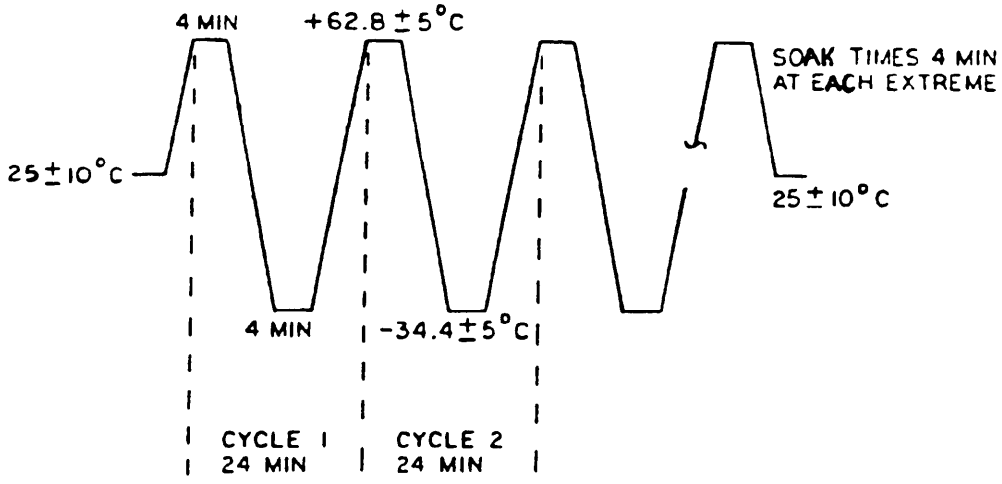


FIGURE 7. Environmental Stress Screen Temperature Profile.

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4.5.3 Cycle frequency. A total of twenty temperature cycles are to be imposed on each unit.

4.5.4 Temperature cycles. Temperature cycling shall be conducted with units nonoperating and without power (static).

4.5.5 Test requirements. After the thermal stress screen, the units shall meet the Group A tests (4.6.2). If a unit is rejected, the unit shall be repaired for the failure (if possible), and be put through one additional cycle of ESS. The unit shall then undergo and pass Group A tests.

4.6 Quality conformance inspection. Quality conformance inspection requirements are defined in 4.6.2 and 4.6.3.

4.6.1 Inspection lot formation. The term "inspection lot" is defined as a homogeneous collection of units of product from which a representative sample is drawn of which is inspected 100 percent to determine conformance with applicable requirements. Units of product selected for inspection shall represent only the inspection lot from which they are drawn and shall not be construed to represent any prior of subsequent quantities presented for inspection. Homogeneity shall be considered to exist provided the inspection lot has been produced by one manufacturer, in one unchanged process, using the same materials and methods, in accordance with the same drawings, same drawing revisions, same specifications and same specification revisions. All material submitted for inspection in accordance with this specification shall comply with the homogeneity criteria specified herein, regardless of the type of inspection procedure which is being applied to determine conformance with requirements. Unless otherwise specified, the "inspection lot" shall be one month production.

4.6.2 Group A tests. Group A tests consist of examinations and performance tests listed in Table II.

4.6.3 Group B tests. Group B tests shall consist of the tests indicated in Table III and shall be conducted on one sample unit per inspection lot which has been subjected to, and passed, Group A tests (see 4.6.2).

4.6.4 Inspection of packaging. The sampling and inspection of the preservation, packing, and container marking shall verify that the units are packaged in accordance with the requirements of 5.1. During the period of first article inspection, one

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Ambient -11.7 25 55 °C

additional unit, packaged in accordance with 5.1, shall be examined to the requirements for quality conformance inspection listed below. Failure of packaging shall be cause for rejection of packaging. Two units per lot shall be examined for the following:

a. Verify that the outside bag is marked in accordance with MIL-STD-129 and is properly closed.

b. Verify that sleeve is around box and that foam properly surrounds inner bag.

c. Verify that the inner bag is properly sealed.

d. Verify that the unit is not damaged after the removal of all packaging materials.

4.6.5 Failure of sample. Should any one sample fail to meet any of the specified examinations and test requirements, acceptance of the product shall be suspended by the Government until necessary corrections to all production since the last test as well as to current production has been accomplished in accordance with the provisions of the contract (see 6.7).

4.7 Inspection equipment.

4.7.1 Government-furnished inspection equipment. Where the contract provides for Government-furnished test equipment, supply and maintenance of test equipment shall be in accordance with the applicable requirements of MIL-I-45607 and MIL-W-63150.

4.7.2 Contractor-furnished inspection equipment.

4.7.2.1 Government design. Unless otherwise specified in the contract, all inspection equipment identified by drawing number in specifications or QAP/QARs forming a part of the contract shall be supplied by the contractor in accordance with the design specified. The contractor may, however, propose alternatives to Government designs (see 6.3).

4.7.2.2 Contractor design. The contractor shall design and supply inspection equipment compatible with the Methods of Inspection specified in 4.8 and 4.9 of this specification and with the component inspection procedures specified in Examination and Test Facilities requirements of MIL-F-13926, whenever Government designs are not specified (see 4.7.2.1). Since tolerance of test equipment is normally considered to be within

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ten percent of product tolerance for which it is intended, this inherent error in the test equipment design must be considered as part of the prescribed product tolerance limit. Thus, concept, construction, materials, dimensions and tolerances used in the design of test equipment shall be so selected and controlled as to ensure that the test equipment will reliably indicate acceptability of a product which does not exceed 90 percent of the prescribed tolerance limit and permit positive rejection when nonconforming. Construction shall be such as to facilitate routine calibration of test equipment. When specified in contract or purchase order, the contractor shall submit inspection equipment designs for government review and evaluation (see 6.3).

4.7.2.3 Test equipment. In conjunction with 4.7.2.1 and 4.7.2.2, the test equipment listed in Table V (Test equipment list) shall be used in the performance of the applicable test methods and procedures as specified in 4.8.

TABLE V. Test Equipment List. 1/

Power Supplies (DC)	Standard - capable of providing levels and limits stated in 4.8.1.
DC Voltmeter/Multimeter	Standard - capable of measuring voltages stated in 4.8.1.
Ammeter/Multimeter	Standard - capable of measuring currents stated in 4.8.1.
Pulse Generator	Standard - capable of providing pulses stated in 4.8.1.
Ohmmeter/Multimeter	Standard - capable of measuring the resistance specified in 4.8.1.5.
Decoder Test Set	Part Number 11835285 - as listed in 4.8.1.3-4.8.1.7.

NOTE:
1/ Must be calibrated to the requirements of MIL-STD-45662 prior to use.

4.8 Methods of inspection. Unless otherwise specified in the contract, the following procedures shall be utilized to

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conduct specified tests. If the contractor chooses to use alternate test procedures, a detailed description of the set-up and procedure shall be submitted to the Government procuring agency for approval prior to use (see 4.7.2.2).

4.8.1 Performance tests. The following tests and examinations shall be performed to demonstrate compliance with 3.4.2. The PWBA shall be connected to the circuit specified in Figure 8. All switches shall be as shown in Figure 8 prior to each test specified in 4.8.1.1 through 4.8.1.7.

4.8.1.1 Current drain. Verify that the quiescent current meets the requirements of 3.4.2.1 by turning ON S8.

4.8.1.2 Rounds count. To verify the rounds count requirements of 3.4.2.2, turn ON S1 momentarily, turn ON S6 and S8, and verify the display reading.

4.8.1.3 Blank fire mode. To verify the blank fire mode requirements of 3.4.2.3, turn ON S8, set S5 to position 1, turn on the pulse generator with pulse width and amplitude set to that shown in Figure 1, turn on the Decoder Test Set AC power with the HIT CODE SELECT thumbwheel set to 23, the NEAR MISS CODE SELECT thumbwheel set to 29 and the COUNTER RESET switch set to AUTO. Turn ON S1 momentarily and verify the HIT WORD COUNTER reads 2, then 32 and the NEAR MISS WORD COUNTER reads 19, then 256. Also verify the LED (DS1 thru DS4) results and the display counts from 75 down to 0 by 1 every 15 seconds. Verify the signals on channels 1 and 2 of the oscilloscope agrees with Figure 2 while the codes are generated, and with S9 set to both positions 1 and 2. Repeat the above with S10 set to position 2.

4.8.1.4 Dry fire mode. To verify the dry fire mode requirements of 3.4.2.4, turn ON S8, turn ON S1 momentarily, turn ON S2, turn on the Decoder Test Set AC power with the HIT CODE SELECT thumbwheel set to 23, the NEAR MISS CODE SELECT thumbwheel set to 29 and the COUNTER RESET switch set to AUTO. Turn on the pulse generator at S5 and set it to produce a pulse of 10 to 60 ms width and 28 $\pm$ .5 V amplitude. Set S5 to position 3, send the pulse to simulate an actual trigger mechanism firing one round, and verify the HIT COUNTER reads 2, then 32 and the NEAR MISS COUNTER reads 19, then 256. Also verify the LEDs (DS1, DS3, and DS4) results. Verify the signals on channels 1 and 2 of the oscilloscope agrees with Figure 2 while the codes are generated, and with S9 set to both positions 1 and 2. Repeat the above with S10 set to position 2.

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4.8.1.5 Test mode. To verify the test mode requirements of 3.4.2.5, turn ON S8, turn on the Decoder Test Set AC power with the HIT CODE SELECT thumbwheel set to 23, the NEAR MISS CODE SELECT thumbwheel set to 29 and the COUNTER RESET switch set to AUTO. Turn ON S1 and S3 momentarily and verify the HIT COUNTER reads 2, then 32, and the NEAR MISS COUNTER reads 19, then 256. Also verify the LEDs (DS1 and DS3) results and the ohmmeter shows at least 1 megaohm across terminals 14 and 15. Verify the signals on channels 1 and 2 of the oscilloscope agrees with Figure 2 while the codes are generated, and with S9 set to both positions 1 and 2. Repeat the above with S10 set to position 2.

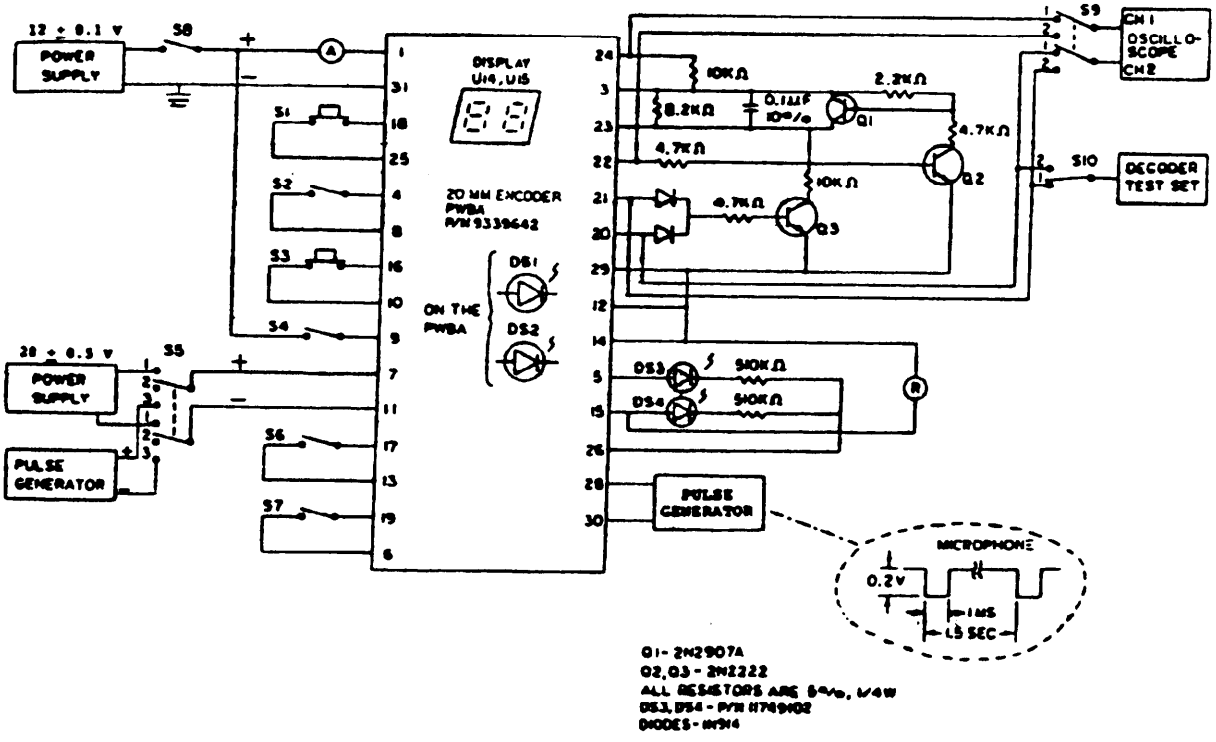


FIGURE 8. Test Circuit for 20mm Encoder.

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4.8.1.6 Kill inhibit mode. To verify the kill inhibit requirements of 3.4.2.6, turn ON S8, turn ON S1 momentarily, replace the Decoder Test Set with an oscilloscope set for a 3 kHz sweep rate, turn ON S2, set S5 to position 1, turn ON S4 and verify that no output appears on the oscilloscope and LED DS2 lights.

4.8.1.7 Boresight mode. To verify the boresight mode requirements of 3.4.2.7, turn ON S8, turn ON S1 momentarily, turn on the Decoder Test Set AC power with the BORESIGHT MODE switch ON, turn ON S4, S5 and S7, and verify the BORESIGHT MODE LED lights and also LED DS2. Verify the signals on channels 1 and 2 of the oscilloscope agrees with Figure 2 while the codes are generated, and with S9 set to both positions 1 and 2. Repeat the above with S10 set to position 2.

4.9 Environmental tests.

a. Unless otherwise specified, environmental tests shall be conducted in accordance with the test conditions and general test performance guidance of MIL-STD-810. Test facilities, chambers and apparatus used in conducting the test required by this specification shall be capable of meeting the conditions specified.

b. Where necessary in the following tests, it shall be permissible to temporarily open chamber doors to permit access to and manipulation of test equipment and sample.

4.9.1 High temperature. This satisfies the storage and operating conditions of high temperature requirements as specified in 3.4.3.

4.9.1.1 Storage and operating. The assembly shall be tested in accordance with MIL-STD-810, Method 501, using the following procedure:

a. Step 1 - Place the test item, configured for high temperature test, into the temperature chamber.

b. Step 2 - Raise the internal temperature to 62.8°C (145°F).

c. Step 3 - Maintain the internal chamber temperature for a period of 48 hours while insuring the relative humidity is not in excess of 15 percent.

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d. Step 4 - Adjust the internal chamber temperature to 55°C (130°F), the highest operating temperature for which the test item is designed to operate, and maintain until temperature stabilization of the item is reached.

e. Step 5 - Operate the test item until the item is stabilized and perform the tests required in 4.8.1.

f. Step 6 - Return the test item, nonoperating, to standard ambient conditions and stabilize.

g. Step 7 - Visually inspect for physical damage and in the absence of physical damage, perform the tests required in 4.8.1.

4.9.2 Low temperature. This satisfies the storage and operating conditions of low temperature requirements as specified in 3.4.3.

4.9.2.1 Storage and operating. The assembly shall be tested in accordance with MIL-STD-810, Method 502, using the following procedure:

a. Step 1 - Place the test item, configured for low temperature test, into the temperature chamber.

b. Step 2 - Lower the internal chamber temperature to -34.4°C (-30°F) and maintain for a period of 24 hours after stabilization.

c. Step 3 - Adjust the internal chamber temperature to -31.7°C (-25°F) and maintain the temperature foa a period of 4 hours.

d. Step 4 - Operate the test item until the item is stabilized and perform the tests required in 4.8.1.

e. Step 5 - Return the test item, nonoperating, to standard ambient conditions and stabilize.

f. Step 6 - Visually inspect for physical damage, and in the absence of physical damage, perform the tests required in 4.8.1.

4.9.3 Shock. The assembly, when mounted on a fixture that

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simulates its next higher assembly (see Figure 9), shall be tested in accordance with MIL-STD-810, Method 516.3, Section II, Procedure I, with the following modifications:

a. Step 3 - the assembly shall be in a non-operational mode.

b. The step 3 test transient shall be in accordance with 3.4.3.3.

c. Step 5 - the assembly inspection and functional verification shall not be conducted until shock has been completed in all three axes.

d. After shock testing the assembly shall not be damaged and shall be subjected to the performance tests specified in 4.8.1.

4.9.4 Vibration. The assembly, when mounted on a fixture that simulates its next higher assembly (see Figure 9), shall be tested in accordance with MIL-STD-810, Method 514.3, Section II, Procedure I, for a duration of 60 minutes per axis with the following modifications:

a. Step 4 - mount the test item on vibration equipment.

b. Step 5 - expose the assembly to test level and durations as specified in 3.4.3.4.

c. Step 6 - does not apply.

d. Step 7 - the assembly inspection and functional verification shall not be conducted until vibration has been completed in all three axes.

e. After vibration testing the assembly shall not be damaged and shall be subjected to the performance tests specified in 4.8.1.

4.10 Certifications. Certification shall be provided that all units have been produced in accordance with the requirements of 3.5 and packaged in accordance with the requirements of 5.1.

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with SPI-P9339642. The level of

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protection shall be A, B, or C, as defined by the procurement document.

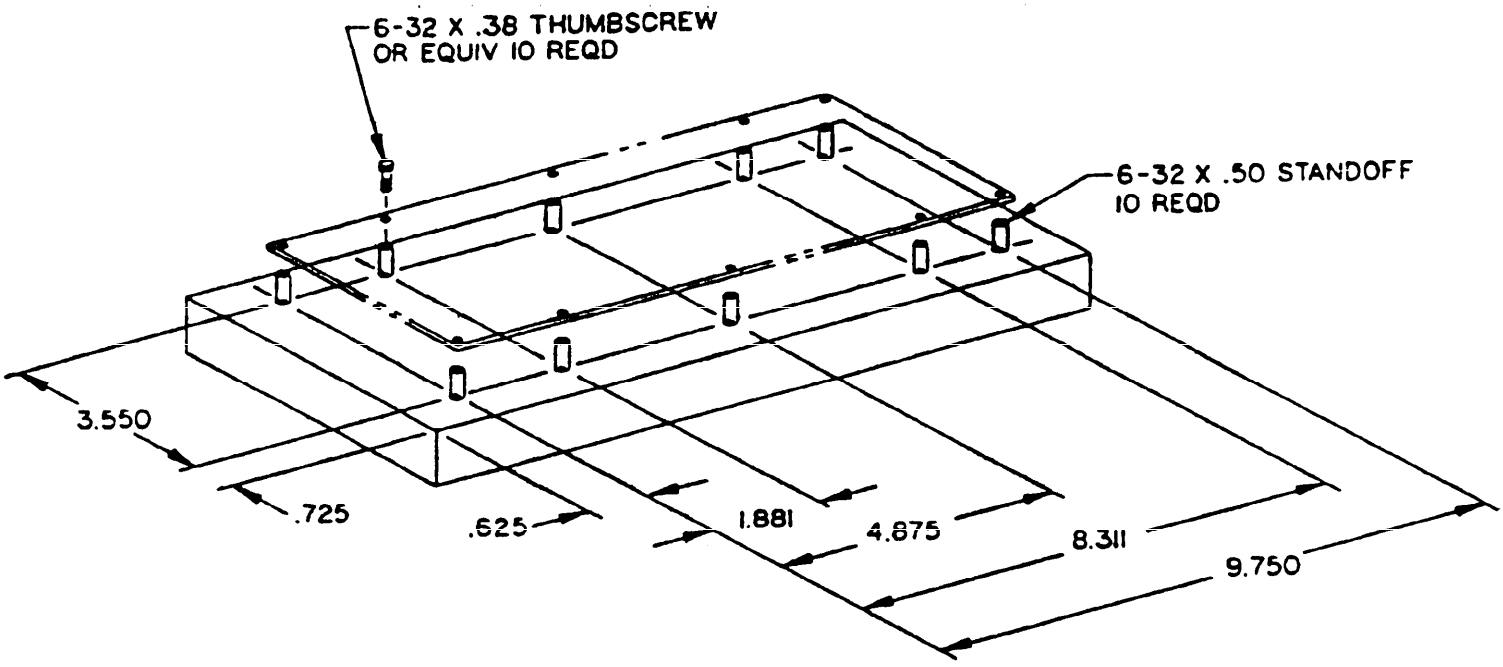


FIGURE 9. Mounting Configuration.

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6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The 20mm Encoder PWBA is part of the Multiple Integrated Laser Engagement System.

6.2 Acquisition requirements. Acquisition documents shall specify the following:

a. Title, number and date of this specification (see page 1).

b. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1).

c. Special tests and examinations required (see 4.2); the type of inspection required; and the activity responsible for testing, and its facility (see 4.1).

d. Whether a first article sample is required (see 3.2), and if so, pertinent details (see 4.4) including disposition of the first article sample.

e. Certification of conformance for each lot or shipment of product.

f. Whether Government-Loaned Property (Government Furnished inspection/test equipment) of 6.8 is available or will the contractor be required to furnish inspection equipment (see 4.7).

g. Applicable level(s) of preservation, packaging, packing and marking instructions (see 5.1).

6.3 Submission of contractor inspection equipment designs for approval. Unless otherwise specified, submit two copies of designs as required to: Commander, U.S. Army AMCCOM, ATTN: SMCAR-QAF-I, Picatinny Arsenal, NJ 07806-5000. This address will be specified on the Contract Data Requirements List, DD Form 1423, in the contract. The letter of submittal shall specify, the contractor; contract number; title, number and revision of this specification; item nomenclature; classification of defect and test paragraph.

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6.4 Definitions.

6.4.1 Hit. Successful transmission of the weapon simulator kill code that results in a "hit" on the target, but that does not "kill" the target. An audio alarm at the target signifies the hit.

6.4.2 Kill. Successful transmission of the weapon simulator kill code that results in the deactivation of or "kill" of the target. An audio alarm at the target signifies the kill.

6.4.3 MILES. Multiple Integrated Laser Engagement System.

6.4.4 Near Miss. Illumination of the target with the near miss laser message resulting in an audio indication to the target that it is under fire.

6.4.5 Round. A laser message that simulates the effects of a round from the weapon.

6.4.6 SME. Standard Measuring Equipment.

6.5 Drawings. Drawings listed in Section 2 of this specification under heading U.S. Army Armament Research, Development and Engineering Center (ARDEC), may also include drawings prepared by, and identified as, U.S. Army Armament Research and Development Command (ARRADCOM), Frankford Arsenal, Rock Island or Picatinny Arsenal drawings. Technical data originally prepared by these activities is now under cognizance of ARDEC.

6.6 Noncompliance. If a sample fails to pass group B inspection, the manufacturer shall notify the qualifying activity and the cognizant inspection activity of such failure and take corrective action on the materials or processes, or both, as warranted, and on all units of product which can be corrected and which were manufactured with essentially the same materials and processes, and which are considered subject to the same failure. Acceptance and shipment of the product shall be discontinued until corrective action, acceptable to the qualifying activity has been taken. After the corrective action has been taken, group B inspection shall be repeated on additional sample units (all tests and examinations, or the test which the original sample failed, at the option of the qualifying activity). Group A inspections may be reinstituted; however, final acceptance and shipment shall be withheld until the group B inspection has shown that the corrective action was successful. In the event of

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failure after reinspection, information concerning the failure shall be furnished to the cognizant inspection activity and the qualifying activity.

6.7 Government-loaned property. Unless otherwise specified (see 6.2), the contracting officer should arrange to loan the following Government-furnished Inspection/Test equipment specified below:

P/N	Title	Ref. Paragraph
11835285	Decoder Test Set	4.8.1.3 - 4.8.1.7

6.8 Subject term (key word) listing.

Fire Control
Multiple Integrated Laser Engagement System
Training Devices

Custodians:
Army - AR

Preparing activity:
Army - AR

(Project 1265-A031)

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INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revisic letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER	2. DOCUMENT DATE (YYMMDD)
	MIL-P-70873 (AR)	10 MARCH 1992

3. DOCUMENT TITLE

4. NATURE OF CHANGE (Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets if needed.)

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME	b. TELEPHONE (Include Area Code)	
US ARMY ARDEC STANDARDIZATION OFFICE	(1) Commercial (201) 724-6674	(2) AUTOVON 880-6674
c. ADDRESS (Include Zip Code) ATTN: SMCAR-BAC-S PICATINNY ARSENAL,NJ 07806-5000	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340	

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